# Daniel Hardesty Lewis

*Forward Deployed Engineer – full-stack, enterprise operations, AI & automation*

Engineer and founder specializing in bridging product engineering and enterprise stakeholders. Built full-stack platforms, automated operational workflows, and shipped customer-facing tooling from scoping through launch.

## Core Expertise

8 years **Full-Stack Engineering**: Build and ship production web applications end-to-end using Python, React, SQL, and Postgres; own from data pipeline through user-facing delivery

4 years **Client Engineering & Onboarding**: Scope requirements, triage requests, and manage technical integrations with enterprise stakeholders across government, financial, and engineering verticals

8 years **Process Automation**: Identify manual bottlenecks and replace them with automated workflows, AI tooling, and repeatable playbooks to scale operations

8 years **Communication & Education**: Translate technical concepts for non-technical audiences; teach, document, and build institutional knowledge across teams and clients

8 years **Technical Leadership**: Drive cross-functional execution across engineering, product, and external partners; mentor engineers through design reviews, code reviews, and instruction

## Technologies

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|---|---|
| Core stack | Python; SQL; TypeScript; React; Postgres; Redis; Elasticsearch; Supabase; FastAPI |
| AI/ML | LangChain; LangGraph; DSPy; RAG; FAISS; PyTorch; scikit-learn; Airflow |
| Data | Pandas; Polars; PyArrow; Parquet; BigQuery; PostGIS; ETL pipelines |
| Infrastructure | Linux; Git; Docker; Kubernetes; Terraform; CI/CD; monitoring & observability |
| Additional | CUDA; C++; Ray; Dask; Spark |

## Experience

### Professional

Apr. 2025 – Present **Co-founder**, *Homecastr*, New York, NY

Consumer real-estate pricing and forecasting platform; owned end-to-end workflow from source data to production web delivery.

- Built a **full-stack web application** from zero to production using React, Python/FastAPI, and Postgres.
- Deployed a semi-supervised VAE on **tabular property records** (tax, sales, permits); achieved 12% holdout error vs. Zillow's off-market Zestimate (8.4%), with leakage-safe evaluation and automated pipelines for annual retraining.
- **Scoped product requirements**, ran user interviews, and iterated on positioning through feedback from a16z, MetaProp, SFR operators, and RE investors.

Nov. 2024 – Present **Co-founder**, *PoliBOM*, Seattle, WA

AI tariff mitigation product for manufacturers, centered on BOM analysis, retrieval, and policy scenario simulation; led **customer-facing scoping, demos, and technical onboarding** for enterprise prospects.

- Specified an **agentic workflow** and schema definitions for BOM parsing, retrieval, and tariff simulation.
- Directed a **conversational AI interface** enabling non-technical decision-makers to retrieve BOMs and receive proactive compliance alerts without engineering support.
- Ran **customer discovery and live demos** with Deloitte, ScoutyAI, and other import/export founders; **scoped integration requirements** for enterprise prospects.
- Contributed to materials presented to KPMG, pre-dating their launch of a tariff-modeling product. Ongoing discovery with ex-Goldman Sachs Prof. Kent Daniel and CBS tariff economists.

Nov. 2023 – Present **Founder**, *Summit Geospatial*, New York, NY

High-resolution elevation data and web delivery for engineering and hazard workflows.

- Engineered a statewide seamless Texas DEM mosaic, resampling 0.5 m LiDAR sources to 1.2 m via nearest-neighbor across 70+ state, federal, and local elevation datasets.
- Built the **web distribution platform end-to-end**: data pipeline, tiling, hosting, and delivery UX.
- Commercial discussions to license elevation datasets to research and AI organizations.

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Aug. 2021 – Aug. 2023 **Senior Data Scientist & Technical Lead**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- **Triaged and scoped enterprise requests** across 6+ state and federal agencies (NOAA, NSF, USACE, GLO), translating technical requirements into RFP and MOU frameworks with clear launch checklists.
- Served as primary **technical point of contact** for the $40M Texas Disaster Information System (TDIS), coordinating between engineering, agency stakeholders, and end-users.
- Built **repeatable processes and documentation** for data onboarding, enabling partner agencies to self-serve integrations with the platform.
- Managed multi-million-dollar compute budgets and orchestrated distributed workloads, identifying opportunities to **automate** pipeline steps and reduce manual intervention.
- **Managed relationships** with federal, state agencies, and universities including NOAA, NSF, USACE, Texas GLO, USC, UT, and A&M.
- **Evaluated and accepted** $10M+ in computing resource requests as Technical Reviewer for Frontera Doctoral Fellows and Large-Scale Computing Pathways, triaging proposals against capacity and impact criteria.
- Co-founded summitgeospatial.com to commercialize platform outputs, owning the customer-facing product from data pipeline through web delivery.

Feb. 2018 – Jul. 2021 **Data Scientist & Research Engineer**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Competed in year-long DARPA World Modelers program to improve disaster resiliency and food security modeling in East Africa, coordinating across distributed engineering and research teams.
- Received recognition from AAAI President Yolanda Gil during her farewell address for contributions to automated scientific modeling.

## Research

May 2025 – Aug. 2025 **Research assistant to Dir. Electrical Engineering Zoran Kostic**, *Columbia University*, New York, NY
Math benchmarking and finetuning of Deepseek's GRPO algorithm.

Jan. 2025 – Oct. 2025 **Industry-sponsored research assistant to Ask2.AI CSO Ali Hirsa**, *Columbia University*, New York, NY

- Developed a multi-asset CVAE latent factor model for **tabular financial data** with Skew-T Mixture priors; achieved 85% $R^2$ vs. commercial SaaS (75%) and Fama-French (58%) on backtested holdout.
- Derived a tractable, axiom-compliant per-feature SHAP algorithm using Hessian-based recursive clustering to provide stable temporal regime-dependent attributions for deep learning models.

July 2019 – Aug. 2019 **Summer research intern to Pres. AAAI Yolanda Gil**, *University of Southern California*, Los Angeles, CA
Integrated hydrological models into MINT's scientific orchestration platform submission to DARPA.

## Teaching

Jan. 2018 – Dec. 2020 **Co-instructor**, *The University of Texas at Austin*, Austin, TX

- Helped design and teach graduate-level courses: *Machine Learning for the Geosciences* for Petrobras geoscientists and petroleum engineers, *Intelligent Systems for the Geosciences*, and *Scientific Computation*, including C++, CUDA, and HPC.

# Select Publications

2021 Gil, Y., et al. (incl. **D. Hardesty Lewis**), "Artificial Intelligence for Modeling Complex Systems: Taming the Complexity of Expert Models to Improve Decision Making". *ACM TiiS*, 2021. DOI: 10.1145/3453172

2019 Garijo, D., et al. (incl. **D. Hardesty Lewis**), "An intelligent interface for integrating climate, hydrology, agriculture, and socioeconomic models". *ACM IUI '19*, 2019. DOI: 10.1145/3308557.3308711

# Education

expected 2026 **M.S., Urban Planning**, *Columbia University*, New York City, NY

2017 **B.S., Pure Mathematics**, *University of Texas*, Austin, TX

2017 **Certificate, Scientific Computation**, *University of Texas*, Austin, TX

# Key Projects

oppsAlert Developed the first property- and owner-level ML model of NIMBYism, using 15+ years of 15,000+ points of individual homeowner protest data provided by the City of Austin.

# Activities

Sailing Volunteer Atlantic SAR of the Cheeki Rafiki

Bikepacking US West & East Coasts

# Languages

Spanish Native Proficiency

French Professional Proficiency

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